

# Online Anchor-Robust Forecasting under Economic Regime Shift

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## Abstract

Most online forecasters handle a regime shift after it happens, by detecting the break and recalibrating. We study forecasters that are robust in advance when the shift travels through an observable driver, such as a geopolitical-risk or policy-uncertainty index. OARF, our method, is online anchor regression: it penalises the part of the forecast error correlated with the driver, and with the penalty at zero it is online gradient descent. An optional layer learns the driver. Because that estimate is sometimes poor, we propose a diagnostic that does not need the true driver. In simulations, reverting to plain online learning on the streams it flags lowers the layer’s mean worst-case error by about a quarter, with a threshold set on the same simulations. On a controlled benchmark OARF more than halves worst-case error relative to online gradient descent. On sixteen years of energy data it is competitive but not significantly better.

*Keywords:* Online learning, Combining forecasts, Robustness, Volatility forecasting, Structural change, Econometric models, Causality

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## 1. Introduction

Economic and financial time series are subject to *regime shifts*: geopolitical shocks, turns in monetary policy, energy crises and, in carbon markets, discrete regulatory interventions such as the activation of the EU Emissions Trading System Market Stability Reserve. Each of these moves the relationship a forecaster is trying to track, so a model that was well calibrated yesterday can be badly miscalibrated tomorrow. The usual response in the online-learning and conformal-prediction literatures is to treat this as a detection problem. One monitors a loss or a coverage statistic, raises an alarm when it drifts, and then recalibrates or re-weights (Gibbs and Candès, 2021; Khan and Syed, 2025; Prinster et al., 2025). This often works, but it has

two weaknesses. The alarm can only fire after the loss has already risen, so the forecaster pays a re-adaptation cost at every break. It also responds to a deterioration in fit, which is a symptom of the shift, and not to whatever produced it.

In many economic settings the shift travels through an *observable channel* of regime drivers, or at least through observable proxies that co-move with those drivers, and the changing distribution of these series is closely associated with what perturbs the forecasting environment. When geopolitical risk spikes or policy uncertainty jumps, the joint distribution of prices, volatilities and their predictors tends to move with it. We treat such series as an observable shift channel and do not claim that they are the structural cause; the strict causal reading is reserved for the synthetic SCM, where it holds by construction. If a forecaster’s residual can be made *invariant* to that channel, meaning that the part of its error that co-moves with the drivers is held at zero, then the forecaster is protected against shifts in the channel’s distribution before they happen, with no detection step. This is the logic of anchor regression (Rothenhäusler et al., 2021), which gives up some in-sample fit in exchange for invariance to interventions on a designated “anchor” variable, and thereby minimises a worst-case risk over bounded interventions. Anchor regression is, however, a batch estimator, and it assumes the analyst already knows which variables make up the anchor. Neither assumption fits a streaming economic deployment, where data arrive one observation at a time and it is uncertain which of the many candidate drivers carries the regime.

This paper addresses the first of these gaps and explores the second. Our main contribution is OARF, a fully streaming anchor-robust forecaster. Its first-order update adds to the ordinary prediction gradient an anchor-robustness term built from exponential-moving-average cross-moments. With the robustness weight set to zero it reduces exactly to online gradient descent, so a single interpretable parameter controls the degree of robustness. Our analysis and claims concern this fixed-channel method. Separately, we sketch an exploratory second layer, OARF-CD, that tries to learn the channel from the stream by identifying the subspace of the observable context along which the relationship to the regime is most unstable. We present it as a diagnostic proof of concept. It is heuristic, our theory does not cover it, and our experiments show that it is unstable across seeds. Since this instability is the practical risk of not knowing the channel, we also ask whether it can be detected online without access to the true channel. A statistic that the discovery layer already computes, the eigengap of its across-block

scatter, tracks retrospective recovery quality. On the synthetic benchmark, falling back to plain online gradient descent on the streams it flags lowers the learned channel’s mean worst-case error by about a quarter. Finally, the usual notions of regret do not describe what such a method is meant to achieve, so we define and analyse *interventional regret*: regret measured against the best regime-conditional predictor under the worst-case bounded future intervention.

*Contributions.* Section 3 gives the method and its theory. OARF implements the anchor-regression objective as a streaming first-order update whose robustness term is built from EMA cross-moments of centred variables. It is an online counterpart of an existing batch estimator, and setting  $\xi = 0$  recovers online gradient descent. The same section describes the channel-discovery rule, which tracks the across-regime nonstationarity of the observable context and returns a nonstationary subspace that need not be causal, and the eigengap diagnostic, which correlates with the learned channel’s recovery of the true subspace (a quantity a deployment cannot observe). We prove that the fixed-channel update is no-regret in the interventional sense against a  $K$ -switch comparator, with bounds of  $O(\sqrt{T(1+K)})$  under oracle anchor moments and  $O((1+K)T^{2/3})$  under online moment estimates. We do not analyse the learned-channel variant. Sections 4 and 5 report the experiments. On a controlled SCM with a held-out intervention grid we measure interventional robustness against ground truth, and use stress tests to find the point at which a noisy or misspecified anchor makes the method worse than using no anchor. We then study forecast combination on sixteen years of public energy data against nine baselines, with Diebold–Mariano (Diebold and Mariano, 1995) and Model Confidence Set (Hansen et al., 2011) tests. The results differ across the two settings. When shifts travel through a clean, observed channel, proactive robustness accepts a higher in-distribution error in exchange for a large reduction in worst-case interventional error. On real energy series, where that channel is weak, the method is competitive with the strongest baselines but not significantly better. Stress tests in Section 5 show where it fails.

## 2. Related work

*Causal robustness and invariance.* The starting point is the observation that a predictor which is *invariant* across environments is also robust to interventions that move between them. Invariant causal prediction (Peters et al.,

2016) formalises this for the identification of causal parents. Anchor regression (Rothenhäusler et al., 2021) turns it into an estimator. By penalising the projection of the residual onto an anchor variable, it interpolates between ordinary least squares and an instrumental-variables solution, and the penalised problem is provably equivalent to minimising worst-case risk over a bounded set of interventions on that anchor. DRIG (Shen et al., 2023) extends the principle to distributional robustness through invariant gradients. All of these are batch procedures that take the environment or anchor variable as given, which is the obstacle for a streaming deployment. OARF keeps the worst-case-intervention interpretation of anchor regression and makes the estimator online. We also study a heuristic add-on that estimates the anchor channel from data. Because that estimate can be unreliable, we pair it with an online diagnostic that flags streams on which the learned channel should not be used; on those streams the forecaster reverts to plain online gradient descent. Our theorem covers only the fixed, analyst-supplied channel. For a deployment that cannot supply one, the diagnostic is a partial safeguard.

*Reactive online adaptation.* A second, very active line of work treats distribution shift as something to detect and then accommodate. Adaptive conformal inference (Gibbs and Candès, 2021) adjusts a miscoverage level online to maintain coverage under drift. Methods from 2025 tighten the detect-then-adapt loop considerably. M-FISHER (Khan and Syed, 2025) maintains an exponential test martingale over non-conformity scores and, when the martingale crosses a Ville threshold, takes a Fisher-preconditioned natural-gradient correction. WATCH/WCTM (Prinster et al., 2025) monitors a weighted conformal test martingale that tolerates benign covariate shift while still flagging harmful concept shift, and adapts the prediction interval accordingly. These are the incumbents against which a proactive method should be judged. OARF tests whether, when an informative anchor is available, the need for reactive triggering can be partly reduced by removing the channel’s influence before it raises the loss.

*Distributionally robust forecasting.* The closest competitors are recent methods that, like OARF, optimise an explicitly worst-case objective. Wang (2026) propose online randomised distributionally robust forecast combination for dependent data. They draw combination weights from a distribution over the simplex and update it by stochastic mirror descent on a Wasserstein ambiguity objective. Liu and Cheng (2026) develop adaptive robust combination with variance- and Expected-Shortfall-scaled exponential weights for

yield-curve forecasting, and Chen et al. (2026) analyse Wasserstein distributionally robust online learning as a primal–dual game with regret guarantees. A parallel strand makes conformal decisions cost-sensitive (Wu et al., 2026). Several of these 2025–2026 works are recent preprints, so we treat them as contemporary literature and competitive baselines, not as settled, peer-reviewed results. They differ from OARF in geometry. They hedge over an *isotropic* ambiguity ball centred at the empirical distribution, which spreads robustness equally in all directions, including directions along which the data never shift. OARF hedges over the anchor-shaped ellipsoid  $\mathbb{E}[aa^\top]$ , so its robustness budget is spent only along the anchor directions in which shifts occur. Our experiments are designed to isolate this difference. They show that the anchor-shaped geometry improves interventional robustness when the anchor is informative, and that the gain shrinks or reverses when it is not.

*Adjacent online and subspace methods.* Nearby work also informs our design. Online invariant and domain-adaptation methods seek predictors that hold across changing environments in a streaming setting. OARF can be seen as an online, single-anchor instance of this idea, specialised to mean-shift interventions, with a closed-form robustness term in place of an explicit environment partition. Nonstationary-subspace and change-adaptive analysis (subspace tracking, stationary-subspace analysis, and the path-length-adaptive online learning of Zhang et al., 2018) is what our discovery layer does: it is an unsupervised nonstationary-subspace estimator, and for that reason we do not call its output causal. What separates OARF from these methods is that it feeds the discovered nonstationary subspace back into an anchor-regression objective whose target is interventional, not purely predictive.

*Online convex optimisation.* The regret analysis rests on dynamic-regret results for online gradient descent (Zinkevich, 2003; Hazan, 2016), the path-length-adaptive refinement of Zhang et al. (2018), and AdaGrad-style preconditioning (Duchi et al., 2011). The economic measurements draw on the daily Geopolitical Risk index (Caldara and Iacoviello, 2022), the Economic Policy Uncertainty index (Baker et al., 2016), realised-volatility modelling in the heterogeneous-autoregressive tradition (Corsi, 2009), and the volatility-timing logic of Fleming et al. (2001).

### 3. The OARF method and its guarantee

*Setup and protocol.* We work in discrete time  $t = 1, \dots, T$ . At each step the forecaster observes predictors  $x_t \in \mathbb{R}^d$  and a vector of candidate regime drivers  $z_t \in \mathbb{R}^p$ , and must predict a scalar target  $y_t$ . In a pure forecasting problem the predictors are explanatory variables; in a combination problem they are the forecasts of several base models. A linear predictor with feature map  $\phi(x) = [1; x] \in \mathbb{R}^{d+1}$  and weights  $w$  emits  $\hat{y}_t = w^\top \phi(x_t)$ , incurs the squared loss  $\ell_t(w) = (y_t - \hat{y}_t)^2$ , and writes its residual  $r_t = y_t - \hat{y}_t$ . The *anchor* is a (possibly learned) linear read-out of the context,  $a_t = B^\top z_t \in \mathbb{R}^q$ , with channel matrix  $B \in \mathbb{R}^{p \times q}$ . The learning protocol is sequential. At step  $t$  we first predict from  $x_t$ , then reveal  $y_t$  and update, and each learner standardises its inputs with past-only statistics. When we later evaluate a frozen learner on held-out interventions, we replay the same frozen standardiser. Section 4 notes one data-preparation step on the real panel that uses full-sample constants.

*Objective.* Following anchor regression (Rothenhäusler et al., 2021), OARF trades predictive fit against invariance of the residual to the anchor by minimising

$$b_{\text{AR}}(w; \xi) = \underbrace{\mathbb{E}[(y - w^\top \phi(x))^2]}_{\text{predictive loss}} + \xi \underbrace{\mathbb{E}[ar]^\top (\mathbb{E}[aa^\top])^{-1} \mathbb{E}[ar]}_{P(w) = \|\text{proj}_a r\|^2}, \quad \xi \geq 0. \quad (1)$$

The penalty  $P(w)$  is the squared length of the projection of the residual onto the anchor. Driving it down forces  $\mathbb{E}[ar] \rightarrow 0$ , so the predictor stops using any component of the signal that co-moves with the regime drivers. At  $\xi = 0$  the objective is ordinary least squares, and as  $\xi$  grows the solution moves toward the invariant predictor. The objective also has a robustness interpretation.

**Proposition 1** (Interventional-robustness equivalence, Rothenhäusler et al., 2021, Thm. 1). *Assume a linear structural causal model in which (a) the anchor  $A$  is a source (exogenous) node; (b) the remaining variables, including the target, follow linear structural assignments  $V = \mathbf{B}V + \varepsilon + \mathbf{M}A$  with  $A \perp \varepsilon$  and  $I - \mathbf{B}$  invertible; (c) the considered interventions are mean-shift interventions  $\text{do}(a := \nu)$  that replace  $A$  by a constant; and (d) the loss is squared error. Then for every  $\xi \geq 0$  there is a radius  $\rho(\xi)$ , given in closed*

form by the cited theorem and strictly increasing in  $\xi$  (with  $\rho(0) = 0$ , the OLS support, and  $\rho(\xi) \rightarrow \infty$  as  $\xi \rightarrow \infty$ , the instrumental-variables limit), such that

$$\min_w b_{\text{AR}}(w; \xi) = \min_w \sup_{\nu: \|\nu\| \leq \rho(\xi)} \mathbb{E}_{\text{do}(a:=\nu)}[(y - w^\top \phi(x))^2]. \quad (2)$$

The equivalence does not hold for arbitrary nonstationarity. It requires the exogenous-anchor SCM (a)–(c) above, and the worst case is taken over mean-shift interventions only. Under these conditions the set hedged over is an ellipsoid shaped by the anchor covariance  $\mathbb{E}[aa^\top]$ , whereas the distributionally robust methods of Section 2 hedge over an isotropic ball; our evaluation tests whether this difference in geometry matters. Our theory inherits the same assumptions (Assumption 2). Where they fail, the stress tests of Section 5 measure the degradation.

*A two-signal example.* Consider two predictors: a stable *parent*  $x_p$  that drives  $y$ , and a *child*  $x_c$  that is generated from  $y$  but is also moved by the anchor,  $x_c = cy + \gamma a + \text{noise}$ . In a calm period both are informative. OLS puts weight on  $x_c$  because it is strongly correlated with  $y$ , and in-distribution this pays off. The child’s predictive value, however, depends on the current anchor level. When the regime moves  $a$  to a new value, the  $\gamma a$  term shifts, the relation between the child and  $y$  breaks, and the OLS forecast deteriorates. Anchor regression detects that the residual of an  $x_c$ -heavy predictor co-moves with  $a$ , penalises it, and shifts weight onto the parent, giving up some calm-period accuracy in exchange. OARF makes this trade online. The synthetic benchmark below uses the same mechanism with several parents and children.

*Streaming update.* The gradient of (1) in the coefficient block separates into a prediction term and a robustness term,

$$\nabla_w b_{\text{AR}} = -2 \mathbb{E}[\phi r] - 2\xi \mathbb{E}[\phi a^\top] (\mathbb{E}[aa^\top])^{-1} \mathbb{E}[ar], \quad (3)$$

and to move from batch to streaming we replace each population moment by an exponential moving average (decay  $\beta$ ) of its sample analogue, computed on centred variables. Subtracting the running means removes the across-regime mean component of the anchor. Under the SCM of Proposition 1, where a regime is a mean shift on  $A$ , this is the component that varies between regimes, and what remains is the within-regime second-moment structure used by the penalty in (1). Centring does not remove the intervention itself, whose effect persists in  $X$  and  $Y$ . It only keeps the slowly varying

regime mean out of the covariance estimate. The channel-discovery layer below makes use of the across-regime mean variation that centring discards. Writing  $\tilde{\cdot}$  for the EMA-centred quantities, and collecting the non-intercept coefficients as  $w_{-0}$ , we maintain running estimates  $m_t^{Xr}$ ,  $m_t^{XA}$ ,  $M_t^{AA}$ ,  $m_t^{Ar}$  of  $\mathbb{E}[\tilde{\phi}\tilde{r}]$ ,  $\mathbb{E}[\tilde{\phi}\tilde{a}^\top]$ ,  $\mathbb{E}[\tilde{a}\tilde{a}^\top]$ ,  $\mathbb{E}[\tilde{a}\tilde{r}]$  (here  $\tilde{\phi}$  stacks the intercept and centred  $\tilde{x}$ , and the intercept row of  $m_t^{XA}$  is held at zero so the penalty never acts on the intercept), and update

$$w_{t+1} = w_t - \eta g_t, \quad g_t = \underbrace{-r_t \phi(x_t)}_{\text{prediction}} - \underbrace{2\xi [0; m_t^{XA}(M_t^{AA} + \epsilon I)^{-1} m_t^{Ar}]}_{\text{anchor robustness}} + \lambda_2 [0; w_{t,-0}]. \quad (4)$$

Of the two factors of  $-2$  in (3), the prediction one is absorbed into the step size  $\eta$  and the robustness one is kept explicit, so that  $\xi$  reads directly as a relative weight. Signs follow gradient descent: the ridge term  $+\lambda_2[0; w_{-0}]$  enters  $g_t$  with a positive sign, and the update  $w \leftarrow w - \eta g_t$  therefore shrinks the non-intercept weights. The ridge  $\epsilon$  stabilises the anchor Gram matrix  $M_t^{AA}$  when it is ill-conditioned (we use  $\epsilon = 10^{-3}$ ). Each step costs  $O(d + pq + q^3)$ , dominated by the  $q \times q$  solve, which is negligible for the small  $q$  we consider. In practice we take the step with AdaGrad per-coordinate scaling (Duchi et al., 2011). The gradient  $g_t$  is exactly (4) and only its step length is preconditioned, so the online-convex-optimisation regret machinery is unchanged, while a single global rate adapts to features on very different scales. With  $\xi = 0$ ,  $g_t$  reduces to the plain squared-loss gradient and the update is online gradient descent; this holds regardless of AdaGrad and standardisation, which act identically on all methods. The decay  $\beta$  controls how quickly the moment estimates forget.

*Online channel discovery.* So far the analyst supplies  $B$ . We now describe a heuristic that estimates  $B$  from the stream. It is not covered by the theorem of the next subsection, and it performs unsupervised nonstationary-subspace estimation, not causal discovery. Under the SCM of Proposition 1, a regime acts on the context by shifting the anchor’s location, so the anchor directions are nonstationary in mean across regimes and stationary nuisance directions are not. We therefore track the across-regime dispersion of the context’s conditional mean and ascend it on a slow timescale  $\eta_B \ll \eta$ ,

$$\max_{\|B\|_F \leq 1} \mathcal{D}(B) = \text{Var}_{\text{regimes}} (\mathbb{E}[B^\top z \mid \text{regime}]) - \gamma_c \|\mathbb{E}[B^\top z]\|^2. \quad (5)$$

This objective selects high-dispersion nonstationary directions. It does not by itself establish that they are exogenous or causal. The grand-mean penalty (second term) discourages loading on the persistent component but does not guarantee exogeneity, and a direction that is only volatile or trending can be selected; the stress tests of Section 5 probe this failure mode. What the layer recovers is the subspace through which regimes appear to act in the observable context, and this coincides with the causal channel only under the SCM assumptions.

Streaming (5) needs a proxy for “regime”. We use a forgetting partition into length- $L$  blocks. A block should be long enough for a stationary driver’s short-horizon autocorrelation to average out within it (so  $L$  should exceed the dominant driver autocorrelation time), and short enough that actual regime changes fall across block boundaries. The block length  $L$  is a hyperparameter selected on the warm-up window. Let  $S$  be the across-block scatter of the block-mean context and  $u$  its grand mean. The ascent direction is  $\nabla_B \mathcal{D} = 2(S - \gamma_c uu^\top)B$ . At each block boundary  $B$  is updated after that step’s robust gradient step and re-orthonormalised onto the Frobenius ball, and the new  $B$  is used from the next step onward. A lightweight non-robust reference predictor is run alongside to keep discovery oriented toward predictively relevant directions. Algorithm 1 gives the complete loop, with discovery as an optional inner block.

*Reliability diagnostic.* The discovery layer can fail silently. Nothing in Algorithm 1 tells the forecaster whether the current  $B$  is a good estimate of the true channel, and in deployment the true channel is unobserved. As a proxy for how well the top- $q$  subspace is identified, we use a quantity the layer already computes. Let  $\lambda_1 \geq \dots \geq \lambda_p$  be the eigenvalues of  $M = S - \gamma_c uu^\top$  at a given block boundary (Eq. (5)), and define the *normalised eigengap*

$$\kappa = \frac{\lambda_q - \lambda_{q+1}}{\sum_{i=1}^p |\lambda_i| + \epsilon} \in [0, 1]. \quad (6)$$

By the Davis–Kahan sin  $\Theta$  theorem, the angle between the top- $q$  eigenspace of an estimated symmetric matrix and that of the population matrix it estimates is controlled by the corresponding population eigengap. A small  $\kappa$  means that the  $q$ -dimensional subspace  $B$  is ascending is only weakly separated from the rest of the spectrum, so which  $q$  directions are selected depends on sampling noise. A large  $\kappa$  means the top- $q$  subspace is well separated and stably identified. The subspace alignment used to score the discovery

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**Algorithm 1** OARF with online channel discovery (one streaming pass)

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- 1: **input:** penalty  $\xi$ , rates  $\eta, \eta_B$ , decay  $\beta$ , ridge  $\epsilon, \lambda_2$ , block  $L$ , channel  $B$   
(or random init if learning)
- 2:  $w \leftarrow 0$ ; EMA means and cross-moments  $\leftarrow 0$ ; block buffer  $\leftarrow 0$
- 3: **for**  $t = 1, \dots, T$  **do**
- 4:   observe  $x_t, z_t$ ; standardise (past-only, winsorised):  $x_s, z_s$ ;  $a \leftarrow B^\top z_s$
- 5:    $\hat{y}_t \leftarrow w^\top [1; x_s]$ ; **emit**  $\hat{y}_t$ ; reveal  $y_t$ ;  $r \leftarrow y_t - \hat{y}_t$
- 6:   update EMA means; centre  $\tilde{x}, \tilde{a}, \tilde{r}$ ; update  $m^{Xr}, m^{XA}, M^{AA}, m^{Ar}$
- 7:   intercept gradient  $g_0 \leftarrow -r$
- 8:   coef. gradient  $g_{-0} \leftarrow -r x_s - 2\xi m^{XA}(M^{AA} + \epsilon I)^{-1} m^{Ar} + \lambda_2 w_{-0}$
- 9:    $w \leftarrow \Pi_{\mathcal{W}}(w - \text{ADAGRADSTEP}(g)) \triangleright g = [g_0; g_{-0}]$ ; ridge sign shrinks  $w_{-0}$
- 10:   **if** learning the channel **then**
- 11:     accumulate  $z_s$  into the block buffer; **every**  $L$  steps update across-block scatter  $S$ , mean  $u$ , and set  $B \leftarrow \text{orth}(B + \eta_B 2(S - \gamma_c u u^\top)B)$
- 12:   **end if**
- 13:   absorb  $(x_t, z_t)$  into the standardiser statistics
- 14: **end for**

Dimensions:  $w \in \mathbb{R}^{d+1}$  ( $w_0$  intercept,  $w_{-0} \in \mathbb{R}^d$ );  $m^{Xr}, x_s \in \mathbb{R}^d$ ;  $m^{XA} \in \mathbb{R}^{d \times q}$  (intercept row excluded);  $M^{AA} \in \mathbb{R}^{q \times q}$ ;  $m^{Ar} \in \mathbb{R}^q$ ; channel  $B \in \mathbb{R}^{p \times q}$ .  $\Pi_{\mathcal{W}}$  projects onto the radius- $D$  ball.

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layer against ground truth in Section 5 needs the true channel, whereas  $\kappa$  is computed entirely from the stream and is available in a live deployment. Davis–Kahan bounds estimation error given a population gap; it says nothing about whether the recovered subspace is exogenous or causal, which still rests on the SCM assumptions of Proposition 1. Section 5 tests whether  $\kappa$  tracks true recovery and whether gating on it helps.

*Interventional regret.* To state the guarantee we need a regret notion that rewards robustness to future interventions as well as tracking of the past. For a radius  $\rho \geq 0$  define the *interventional regret*

$$\text{IntReg}_T(\rho) = \sum_{t=1}^T \ell_t(w_t) - \min_{u \in \mathcal{W}} \sum_{t=1}^T \sup_{\|\nu_t\| \leq \rho} \mathbb{E}_{\text{do}(a:=\nu_t)} [(y_t - u_t^\top \phi(x_t))^2] : \quad (7)$$

the learner’s realised loss, measured against the best comparator in a class  $\mathcal{W}$  that is itself stress-tested by the worst bounded intervention at every step.

*Choice of regret notion.* Ordinary dynamic regret measures the learner against the best predictor on the realised sequence. It rewards fitting whatever just happened, which is the reactive behaviour we want to avoid, and says nothing about robustness to a shift that has not yet occurred. Suppose two regimes differ only in the anchor’s location. A predictor that exploits an anchor-correlated signal can have low dynamic regret, because it tracks each realised regime well, and still incur a large loss as soon as the anchor moves to a new value. Interventional regret requires the comparator to do well under the worst bounded intervention, so the learner can compete only by being robust across the whole intervention class. The comparator is stress-tested while the learner is scored on realised loss. This asymmetry favours the comparator and makes the bound harder to attain. The realised stream is itself one admissible intervention (Assumption 2), so the learner is implicitly stressed by the shifts it actually meets. One could instead score learner and comparator under the same counterfactual intervention path. We do not, because the learner makes its online decisions on the realised stream and the counterfactual grid is used only for evaluation. Equation (7) is therefore a *robust-comparator* regret (realised loss for the learner against a worst-case-stressed comparator), not a fully counterfactual online regret in which the learner also receives counterfactual feedback. The guarantee does not imply superiority under arbitrary adversarial shifts, since the comparator class is regime-conditional. It gives no guarantee for the learned channel, and it does not imply low realised MSE when the anchor is wrong; the stress tests and scenarios show that case explicitly.

The guarantee rests on the following assumptions.

**Assumption 1.** (*Boundedness.*) *The standardised, winsorised features and anchor are bounded, so the per-step gradients of the surrogate below are bounded in norm by a constant  $G$ , and the comparator class  $\mathcal{W}$  lies in a Euclidean ball of radius  $D$ .*

**Assumption 2.** (*Linear-SCM anchor model.*) *The data follow a linear structural causal model in which the anchor enters as in Section 3, so the equivalence of Proposition 1 holds within each regime, and the radius is adequate,  $\rho(\xi) \geq \max_t \|\nu_t^*\|$ , where  $\nu_t^*$  is the realised regime’s anchor displacement (so each realised regime is one admissible intervention in the ball).*

**Assumption 3.** (*Oblivious regime schedule and comparator.*) *The regime schedule is exogenous and oblivious to the learner; the per-step worst-case*

risk  $\mathcal{R}_t$  defined below is therefore a deterministic function of  $w$ , and  $\mathcal{W}$  is the class of regime-conditional comparators that switch at most  $K$  times (path length  $P_T \leq 2DK$ ).

**Assumption 4.** (*Within-regime stationarity.*) Within each regime the joint distribution of  $(\phi, a, y)$  is stationary with mixing fast relative to the EMA window, so the EMA moment estimates obey the concentration of Lemma 3.

We view Algorithm 1 as online gradient descent fed inexact gradients on the convex per-step worst-case risks  $\mathcal{R}_t(w) := \sup_{\|\nu\| \leq \rho(\xi)} \mathbb{E}_{\text{do}(a:=\nu)}[(y_t - w^\top \phi(x_t))^2]$  of (7). The full argument, in three lemmas, is given in Appendix Appendix A; here we state the result and sketch the proof.

**Theorem 1** (Fixed-anchor OARF interventional regret). *Under Assumptions 1–4, Algorithm 1 with a fixed channel and the Ader step wrapper attains sublinear expected interventional regret against any  $K$ -switch comparator, with no prior knowledge of  $K$ :*

- (i) *with oracle anchor second moments,  $\mathbb{E}[\text{IntReg}_T(\rho(\xi))] = O(\sqrt{T(1+K)})$ ;*
- (ii) *with the EMA moment estimates of Algorithm 1 and decay  $1 - \beta_T = T^{-2/3}$ ,  $\mathbb{E}[\text{IntReg}_T(\rho(\xi))] = O((1+K)T^{2/3})$ .*

*Proof sketch (full proof in Appendix Appendix A).* (1)  $\mathcal{R}_t$  is convex and, by Proposition 1, equals the anchor-penalised risk within a regime; the update direction is  $\nabla \mathcal{R}_t(w_t) + \zeta_t$ , where  $\zeta_t$  splits into a zero-mean martingale part  $M_t$  (one-sample gradient noise) and a deterministic part  $b_t$  (EMA moment-estimation error). (2) The path-length-adaptive Ader wrapper (Zhang et al., 2018) gives dynamic regret  $O(\sqrt{T(1+K)})$  against a  $K$ -switch comparator without knowing  $K$ , plus the inexactness term  $\sum_t \langle \zeta_t, w_t - u_t \rangle$ . (3) Against the deterministic (oblivious-schedule) comparator,  $\mathbb{E} \sum_t \langle M_t, w_t - u_t \rangle = 0$ , while  $D \sum_t \mathbb{E} \|b_t\| \leq D(c_1 T \sqrt{1-\beta} + c_2 K / (1-\beta))$  is zero with oracle moments and  $O((1+K)T^{2/3})$  at  $1 - \beta_T = T^{-2/3}$ . Bounding realised loss by the worst-case risk,  $\mathbb{E}_t[\ell_t(w_t)] \leq \mathcal{R}_t(w_t)$  (Assumption 2), gives the two rates; both are  $o(T)$ .  $\square$

*Remark 1* (Reading the two rates). The gap between (i) and (ii) is the cost of learning the robustness geometry online without changepoint detection. A detection-and-reset estimator could reduce the moment error at the  $O(1/\sqrt{n})$  within-regime rate and recover (i), but it would reintroduce the triggers

OARF is designed to avoid. The  $T^{2/3}$  rate is what a proactive, exponentially forgetting estimator pays, and it is still sublinear. The theorem applies to the Ader-wrapped version, while the experiments use a single AdaGrad instance, so the algorithm we run is not exactly the one we analyse. AdaGrad keeps per-coordinate adaptive steps but drops the  $K$ -adaptivity of the Ader wrapper, which is what makes the bound hold for an unknown number of switches. We use it because it is simpler, has fewer tuning parameters and is more robust on heavy-tailed streams. The cost is that the worst-case  $K$ -switch guarantee is established only for the Ader variant. In a pilot on the synthetic benchmark (Appendix Appendix D, 12 seeds) the two step rules give qualitatively similar results. Both cut the worst-case interventional error far below OGD, their per-seed worst-case errors are strongly rank-correlated (Spearman  $\rho = 0.80$ ), and AdaGrad is slightly more accurate and stable. We therefore report AdaGrad as a simpler stand-in.

*Remark 2* (Scope of the guarantee). First, risk-adjusted no-regret is attainable only under stochastic, stationary or ergodic data, and not against a fully adversarial environment (Uziel and El-Yaniv, 2018). For this reason  $\mathcal{W}$  is regime-conditional, and observable regime drivers must enter the statement, since they define the regimes the comparator may condition on. Second, anchor-based methods deliver interventional robustness, not causal identification. The estimand is the diluted-causal parameter, and we do not claim to recover a structural coefficient.

#### 4. Experimental design

We evaluate OARF in two settings. A synthetic structural causal model provides ground truth: we can intervene on the anchor by construction and so measure interventional robustness directly. A sixteen-year study on real economic series then tests whether the gains carry over to data with no ground-truth intervention and a milder regime structure.

*Synthetic anchor-SCM.* The synthetic generator (Table 1) is built so that the gap between ordinary prediction and interventional robustness can be measured. An exogenous anchor  $A$  and a hidden confounder  $H$  drive a block of causal parents  $X_{\text{par}}$  of the target, which is generated as  $Y = X_{\text{par}}^\top b_{\text{par}} + H^\top d_H + \varepsilon_Y$ . The mean of  $A$  shifts across eight regimes, and these shifts are the in-sample interventions. A second block of children is generated downstream of the target,  $X_{\text{ch}} = Y c_{\text{ch}} + A W_{Ac} + H W_{Hc} + \varepsilon_{\text{ch}}$ , so the children are

anti-causal predictors that also load on the anchor. In sample the children are highly informative about  $Y$ , since they are generated from it, and ordinary least squares and online gradient descent put heavy weight on them. A child’s relationship to  $Y$  is, however, contaminated by the anchor term  $AW_{Ac}$ . Under an intervention  $\text{do}(A := \nu)$  this term is set externally, the child becomes misleading, and the error it injects grows with  $\|\nu\|$ . Only a predictor that uses the parents alone is unaffected by the intervention, and this is the gap that anchor robustness is meant to close. To give the discovery layer something to recover, the observable context  $z$  is a random rotation of the anchor together with  $p - q$  stationary AR(1) decoy drivers, so there is a ground-truth channel  $B_{\text{true}}$  with  $ZB_{\text{true}} = A$ . We hold out a fixed grid of interventions (ten magnitudes, six directions, four hundred draws each) that extends to anchor values beyond the training range. Every interventional number in Section 5 is computed on this grid.

Table 1: Configuration of the synthetic anchor-SCM. Unless otherwise stated, the main performance summary (Table 5) and the channel-discovery figure aggregate over 50 seeds; the component ablations (Table 6) use 16 seeds and the assumption-violation and selector experiments (Tables 7–8) use 8 seeds. Each table and figure states its own seed count.

| Symbol                          | Meaning                                             | Value    |
|---------------------------------|-----------------------------------------------------|----------|
| $T$                             | stream length                                       | 8000     |
| $d_{\text{par}}, d_{\text{ch}}$ | causal parents / anti-causal children               | 3, 3     |
| $p$                             | observable context dimension                        | 8        |
| $q$                             | true anchor (channel) dimension                     | 2        |
| $h$                             | hidden-confounder dimension                         | 2        |
| #regimes                        | anchor-mean regimes ( $\Rightarrow$ 7 changepoints) | 8        |
| $\sigma_{\text{shift}}$         | across-regime anchor-mean scale                     | 1.5      |
| $\sigma_A$                      | within-regime anchor sd                             | 0.6      |
| $c_{\text{ch}}$ scale           | strength of the target $\rightarrow$ child edge     | 1.3      |
| $W_{Ac}$ scale                  | anchor $\rightarrow$ child contamination            | 1.1      |
| confounding                     | strength of the $H \rightarrow (X, Y)$ path         | 0.9      |
| $\sigma_X, \sigma_Y$            | predictor / target noise                            | 0.4, 0.5 |

*Real-data panel and forecasting task.* For the real study we assemble a public daily panel covering 2010–2026 (Table 2), aligned to a business-day calendar with capped forward-fill. The targets are Brent crude and Henry Hub natural gas, and the task is the one-step-ahead forecast of a *daily realised-variance*

*proxy*. With daily closing prices only, true intraday realised variance cannot be formed. The target is therefore the log of a five-day rolling mean of squared daily log-returns, a coarse variance proxy at daily frequency and not an intraday RV estimator. We forecast variance instead of returns because daily returns are close to unforecastable, so a return study would mostly measure noise. The variance proxy is predictable through volatility clustering and depends on the regime, which is where regime robustness can matter. The macro drivers (the CBOE VIX, the dollar, the ten-year yield, and the daily Geopolitical Risk and Economic Policy Uncertainty indices) form the observable context. VIX is equity-index implied volatility, and we use it only as a cross-asset uncertainty driver, not as a measure of commodity implied volatility. A commodity-specific implied-volatility series would be preferable but is not freely available at daily frequency over this span. The levels of the three uncertainty indices are the hand-chosen anchor for the fixed-channel variant, while OARF-CD learns its own.

Table 2: The public real-data panel. All series are free and citation-requested and are fetched by the released download script.

| Series             | Source / ticker                                   | Span (daily) | Role              |
|--------------------|---------------------------------------------------|--------------|-------------------|
| Brent crude        | FRED DCOILBRETEU                                  | 2010–2026    | target (log varia |
| Henry Hub gas      | FRED DHHNGSP                                      | 2010–2026    | target (log varia |
| VIX                | FRED VIXCLS                                       | 2010–2026    | driver / anchor   |
| EUR/USD            | FRED DEXUSEU                                      | 2010–2026    | driver            |
| 10y Treasury       | FRED DGS10                                        | 2010–2026    | driver            |
| Geopolitical Risk  | Caldara–Iacoviello (Caldara and Iacoviello, 2022) | 2010–2026    | driver / anchor   |
| Policy Uncertainty | Baker–Bloom–Davis (Baker et al., 2016)            | 2010–2026    | driver / anchor   |

*Forecast-combination pool.* In the real study every method combines the same pool of one-step-ahead volatility forecasts, made up of standard models. The pool contains the heterogeneous autoregressive (HAR) cascade of daily, weekly and monthly lagged variance (Corsi, 2009); an exponentially weighted RiskMetrics estimator; a GARCH(1, 1) conditional variance; (log) VIX as a cross-asset uncertainty signal; and a regularised regression on these together with the macro drivers. The members capture different features of volatility. HAR encodes long-memory persistence, the EWMA and GARCH terms capture short-run clustering with different decay structures, and the VIX signal adds cross-asset information that the autoregressive models miss.

No single member dominates across regimes, which is why combining them is worthwhile, and this is the setting for which the 2026 distributionally robust combiners were proposed. Each member forecasts day  $t$  from data through  $t - 1$ ; the protocol paragraph below notes one qualification for the regularised regression.

*Baselines.* We compare against eleven methods, grouped by the question each answers about OARF (Table 3). The *basic baselines* are online gradient descent (which is OARF with  $\xi = 0$ ), a rolling-window ridge regression, and adaptive conformal inference (Gibbs and Candès, 2021). They show the accuracy attainable with no robustness mechanism and with a strong batch refit, and so calibrate both ends of the scale. The 2025 reactive methods, M-FISHER (Khan and Syed, 2025) and WATCH/WCTM (Prinster et al., 2025), follow the detect-then-adapt approach that OARF is designed to replace. Comparing against them tests whether removing a channel’s influence in advance does better than reacting to it after the loss rises. The 2026 distributionally robust methods are randomised Wasserstein combination (Wang, 2026), the variance- and ES-scaled exponential-weight combiners (Liu and Cheng, 2026), Wasserstein online learning (Chen et al., 2026) and cost-aware conformal inference (Wu et al., 2026). They are the closest comparison, because they also optimise a worst-case objective. They differ from OARF mainly in the shape of the set they hedge over, an agnostic ball in place of an anchor-shaped ellipsoid, so this comparison bears on the value of the anchor geometry. Finally, the batch causal-robust estimators, anchor regression (Rothenhäusler et al., 2021) and DRIG (Shen et al., 2023), use the same invariance principle as OARF, but offline and with the channel supplied. They are offline references. Anchor regression, which is given the true anchor, is the natural best case for the fixed-channel online method. All methods are run under the same protocol, with AdaGrad-stabilised steps where applicable. Appendix Appendix C gives, for each baseline, the loss it optimises, its update rule, its output type, whether it is online, rolling or batch, and whether it uses the anchor.

*Forecast output.* Every method produces a one-step point forecast and is scored by the same squared loss under the same protocol. The linear learners (OARF, OARF-CD, OGD, Rolling-OLS, W-DRO-OL) and the combination methods (DR-Combo, FC-DRO/-ES) output  $w_t^\top \phi(x_t)$  directly. The conformal methods use a quantile read-out. ACI and CostACI maintain

Table 3: The eleven comparison methods, organised by the question each answers about OARF. The column “Hedges over” names the uncertainty geometry; OARF is the only method whose set is an anisotropic anchor-shaped ellipsoid, and the only proactive (non-triggered) robust learner.

| Method                         | Year    | Family                 | Hedges over                            | Posture             |
|--------------------------------|---------|------------------------|----------------------------------------|---------------------|
| <b>OARF</b> (+ OARF-CD, expl.) | ours    | anchor-robust OCO      | anchor ellipsoid $\mathbb{E}[aa^\top]$ | proactive           |
| OGD                            | 2003    | basic baseline         | –                                      | –                   |
| Rolling-OLS                    | –       | basic baseline         | –                                      | windowed            |
| ACI                            | 2021    | conformal              | coverage level                         | reactive            |
| M-FISHER                       | 2025    | test-time adaptation   | – (martingale trigger)                 | reactive            |
| WATCH/WCTM                     | 2025    | conformal martingale   | interval width                         | reactive            |
| DR-Combo                       | 2026    | robust combination     | Wasserstein ball                       | mirror descent      |
| FC-DRO(-ES)                    | 2026    | robust combination     | variance / ES                          | exponential weights |
| W-DRO-OL                       | 2026    | robust online learning | Wasserstein ball                       | primal–dual         |
| CostACI                        | 2026    | cost-aware conformal   | coverage level                         | reactive            |
| AnchorReg / DRIG               | ’21/’23 | causal (batch)         | anchor / environments                  | windowed            |

5/50/95% pinball heads and take the median head as the point forecast (their adaptive level affects only the interval), and WATCH/WCTM uses an OGD point head with a martingale-driven interval. Only these three also report coverage, CRPS and width. M-FISHER uses its adapted point head. All methods are therefore comparable on point loss, and the interval-based ones are also compared on distributional metrics where these are defined.

*Metrics, protocol and hyper-parameters.* We report point accuracy (overall and root MSE; per-regime, worst-regime and post-changepoint MSE over 50-step windows), interventional robustness on the held-out grid (the curve  $\text{MSE}(\nu)$  and the worst-case error), and coverage, CRPS and interval width for the quantile heads. As a secondary check (Appendix) we measure the economic value of the volatility forecasts with a vol-timing strategy (Fleming et al., 2001). Significance is assessed with the pairwise Diebold–Mariano test (Diebold and Mariano, 1995) and the Model Confidence Set (Hansen et al., 2011). The protocol uses an expanding window: the first fifth of each series is a warm-up and hyper-parameter block, and the remainder is evaluated. The main synthetic summary aggregates over 50 seeds (medians with interquartile ranges), the ablations over 16 seeds and the scenario and selector experiments over 8 seeds; each table states its own count. Real-data numbers are averages over three rolling origins. The real panel ends on 22 June 2026, the 2026 data are complete up to that date, and the evaluation window is the post-warm-up remainder. Real-data regimes are terciles of a volatility-stress index  $s_t$ , the lagged 20-day mean of the standardised log variance proxy, split at its

full-sample 33/67% quantiles into calm, normal and turbulent states. The labels are used only to score forecasts and are never seen by any method; the regime-aware metrics also cover the transitions between these states. Results are qualitatively unchanged under a two-state median split. All hyper-parameters are listed in Table 4 and were fixed a priori or selected once on the warm-up block. The held-out intervention grid is used only for evaluation, never to tune  $\xi$  or any other parameter.

One data-preparation step on the real panel uses full-sample information: the driver series are centred and scaled with their full-sample mean and standard deviation before they enter the stream. For the learners this makes no difference after the first few warm-up observations, because each learner re-standardises its inputs with past-only statistics, and that transform does not change under an affine rescaling of its input. It does affect the regularised-regression member of the forecast pool, whose ridge penalty acts on the pre-scaled drivers, so the strength of its penalty uses full-sample scale information. This is a mild departure from a strictly past-only protocol, and the reported runs include it. Because every method combines the same pool, it affects all methods alike. Parts of the analysis code were written with the help of an AI coding assistant (see the acknowledgments); all of it is public and regenerates every reported number.

Table 4: Hyper-parameters. Values were fixed a priori or chosen once on the warm-up window; none were tuned on the evaluation period or the intervention grid.

| Symbol        | Meaning                               | Value                     |
|---------------|---------------------------------------|---------------------------|
| $\eta$        | learning rate (AdaGrad base)          | 0.1                       |
| $\xi$         | robustness penalty (headline OARF)    | 6 (synthetic), 1.5 (real) |
| $\beta$       | EMA decay for moments                 | 0.98                      |
| $\epsilon$    | ridge on the anchor Gram              | $10^{-3}$                 |
| $\lambda_2$   | $\ell_2$ weight                       | $10^{-4}$                 |
| $\eta_B$      | channel-discovery rate                | 0.3                       |
| $L$           | discovery block length                | 40                        |
| $\gamma_c$    | grand-mean penalty (discovery)        | 0.3                       |
| $q$           | channel dimension                     | 2 (synthetic), 3 (real)   |
| winsorisation | standardised-feature clip             | $\pm 8$                   |
| $\alpha$      | nominal miscoverage (conformal heads) | 0.1                       |
| warm-up       | fraction reserved for warm-up/HP      | 0.2                       |

## 5. Results

The experiments below use between 8 and 50 seeds, and some report medians while others report means, so the absolute worst-case error of the same method differs between tables. Comparisons are meant within a table.

*Synthetic benchmark.* Table 5 and Figures 1–4 report the controlled benchmark over 50 seeds, as medians with interquartile ranges. The in-distribution and interventional rankings are nearly reversed. In distribution, where every regime has been seen, the batch and reactive methods are the most accurate: rolling-window OLS, DRIG and the conformal heads have the lowest median overall MSE ( $\approx 0.011$ ). Fixed-channel OARF does not use the anchor-correlated child signal and pays for this with a higher median in-distribution error (0.019). The difference is larger than rounding and is a real cost. Under intervention the order reverses. OARF attains a median worst-case interventional MSE of 0.041, about 57% below online gradient descent (0.095), 58% below the Wasserstein-DRO online learner (0.098) and WATCH/WCTM (0.099), and 80% below the reactive M-FISHER (0.204). It comes close to the batch anchor-regression reference (0.026) without matching it, and that reference is given the true anchor. The interquartile ranges overlap because the interventional metric is heavy-tailed and seed-dependent, so we read the gap as a robust shift in the median and not as dominance on every seed. Figure 6 shows the relationship between alignment and robustness seed by seed.

As the intervention magnitude grows, OARF and the batch reference stay nearly flat while the reactive and agnostic-robust methods climb steeply (Figure 3a). Sweeping  $\xi$  traces the trade-off between adaptation and robustness (Figure 3b). The update acts directly on the anchor-residual correlation: OARF holds it near zero, whereas for OGD it spikes at every regime change (Figure 4).

The learned-channel variant is weaker and less reliable. OARF-CD recovers the true subspace on average (alignment  $0.88 \pm 0.16$ , Figure 5), and its median interventional MSE (0.049) is close to that of fixed-channel OARF. Its distribution is heavy-tailed, however. The worst-case interventional MSE averages 0.147 across seeds, roughly three times the median, and on a non-trivial fraction of seeds the discovered channel is poor enough that the error approaches or exceeds that of OGD (median 0.095), which applies no anchor penalty. Figure 6 shows this gap between median and mean; we examine its source below.

Table 5: Synthetic anchor-SCM over 50 seeds (lower is better). For the overall and worst-do( $A$ ) columns we report the median with the interquartile range  $[q_{25}, q_{75}]$  in brackets; the worst-regime and post-changepoint columns give the median only. Medians and IQRs are robust to the heavy right tail of the interventional metric. The final column is the held-out interventional metric, and the best fully online method is in bold. AnchorReg and DRIG are batch methods, and AnchorReg is also given the true anchor, so both serve as references rather than competitors. The interventional median of OARF-CD is competitive but its tail is heavy (mean 0.147); see the text.

| Method                                       | overall MSE          | worst-regime MSE | post-cp MSE | worst-do( $A$ ) MSE         |
|----------------------------------------------|----------------------|------------------|-------------|-----------------------------|
| <b>OARF</b> (main, fixed channel)            | 0.019 [0.010, 0.043] | 0.030            | 0.021       | <b>0.041 [0.015, 0.071]</b> |
| OARF-CD ( <i>exploratory</i> , learned $B$ ) | 0.018 [0.010, 0.040] | 0.029            | 0.021       | 0.049 [0.020, 0.124]        |
| OGD                                          | 0.012 [0.008, 0.033] | 0.022            | 0.018       | 0.095 [0.031, 0.220]        |
| Rolling-OLS                                  | 0.010 [0.005, 0.030] | 0.015            | 0.012       | 0.083 [0.025, 0.278]        |
| ACI                                          | 0.011 [0.006, 0.030] | 0.015            | 0.013       | 0.071 [0.022, 0.164]        |
| M-FISHER (2025)                              | 0.021 [0.016, 0.043] | 0.043            | 0.039       | 0.204 [0.072, 0.387]        |
| WATCH/WCTM (2025)                            | 0.012 [0.008, 0.033] | 0.022            | 0.018       | 0.099 [0.031, 0.220]        |
| W-DRO-OL (Chen 2026)                         | 0.012 [0.007, 0.033] | 0.022            | 0.018       | 0.098 [0.033, 0.288]        |
| CostACI (2026)                               | 0.011 [0.006, 0.030] | 0.015            | 0.013       | 0.071 [0.022, 0.164]        |
| AnchorReg (batch, true anchor)               | 0.011 [0.005, 0.031] | 0.014            | 0.011       | 0.026 [0.010, 0.074]        |
| DRIG (batch)                                 | 0.010 [0.005, 0.030] | 0.015            | 0.014       | 0.116 [0.032, 0.281]        |

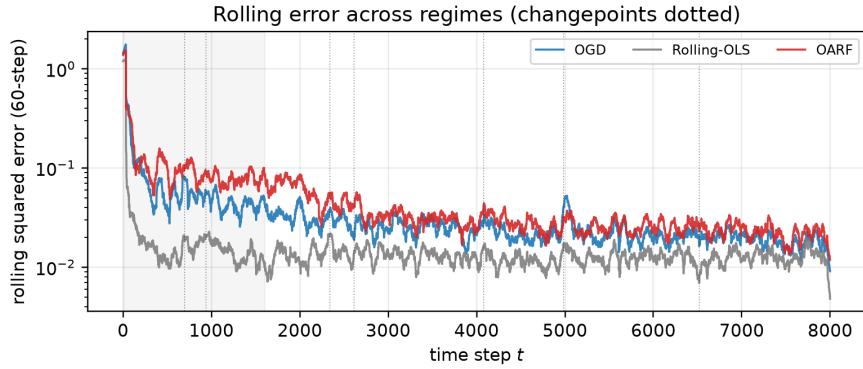


Figure 1: Rolling squared error across regimes on a representative seed (changepoints dotted, the shaded band is the warm-up window). The reactive baselines spike at every break; OARF stays comparatively flat.

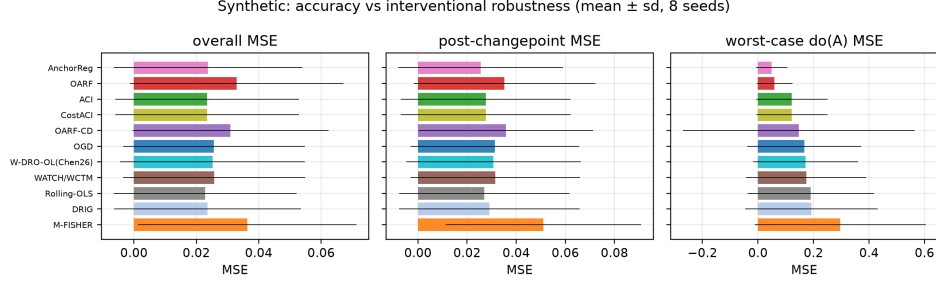
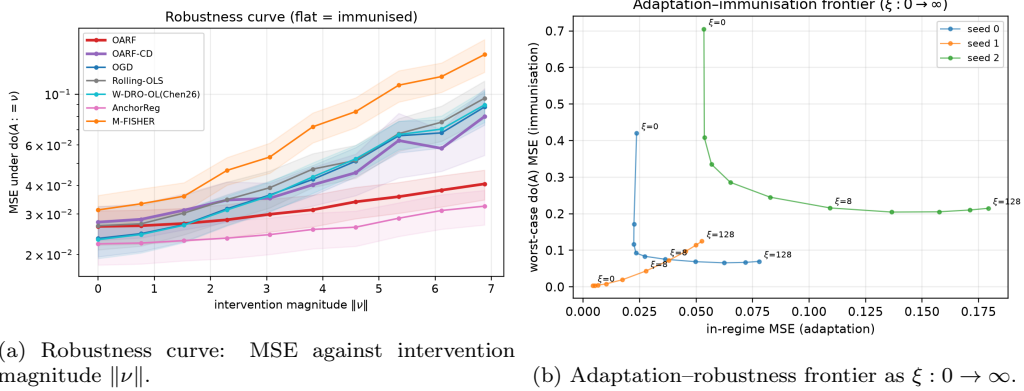


Figure 2: Overall, post-changepoint and worst-case interventional MSE on the synthetic benchmark, sorted by interventional robustness. The reactive and batch methods win in distribution and OARF and the batch oracle win under intervention, so the two panels are nearly mirror images.



(a) Robustness curve: MSE against intervention magnitude  $\|\nu\|$ .

(b) Adaptation-robustness frontier as  $\xi : 0 \rightarrow \infty$ .

Figure 3: (a) OARF and the batch oracle remain near-flat as interventions grow, while reactive and agnostic-robust methods degrade steeply. (b) The penalty  $\xi$  moves the learner continuously from the online-gradient-descent corner to maximal robustness.

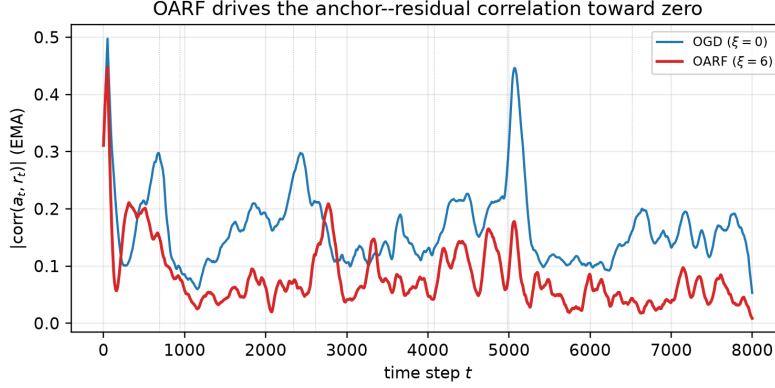


Figure 4: Anchor-residual correlation. The smoothed correlation  $|\text{corr}(a_t, r_t)|$  is held near zero by OARF ( $\xi = 6$ ) but spikes for online gradient descent ( $\xi = 0$ ) at each regime change (dotted).

*Learned channel.* When it has to find its own anchor, OARF-CD recovers the true subspace on average. Mean principal-subspace alignment is  $0.88 \pm 0.16$ , against a random-subspace baseline of 0.42, and individual seeds climb toward 0.99 as the stream lengthens (Figure 5). High mean alignment does not give stable robustness, though. A seed’s final alignment is only noisily related to its worst-do( $A$ ) MSE (Figure 6), and the seeds with transiently poor alignment account for the heavy interventional tail in Table 5. The layer therefore recovers a nonstationary subspace but does not yet match the reliable interventional robustness of the fixed channel. It is a proof of concept for removing the analyst-supplied anchor, not yet a dependable substitute for a known channel, and this tail is the main practical limitation of the discovery layer.

*Online reliability diagnostic.* Figure 6 shows that the failures of the discovery layer are a discrete tail risk. Most seeds recover a well-aligned channel, a minority do not, and those seeds dominate the mean worst-case error. If they could be flagged while the stream runs, without consulting the true channel (which a deployment never observes), the layer could decline to trust its own channel on those streams. Figure 7a tests whether the eigengap  $\kappa$  of Eq. (6) can serve as such a flag. Across the 50 seeds,  $\kappa$  correlates with the true subspace alignment, which is otherwise unobservable, at  $r = 0.62$ , and 13 of the 16 seeds whose final alignment falls below 0.8 also fall in the bottom quartile of post-warm-up eigengap. Figure 7b uses this as a gate.

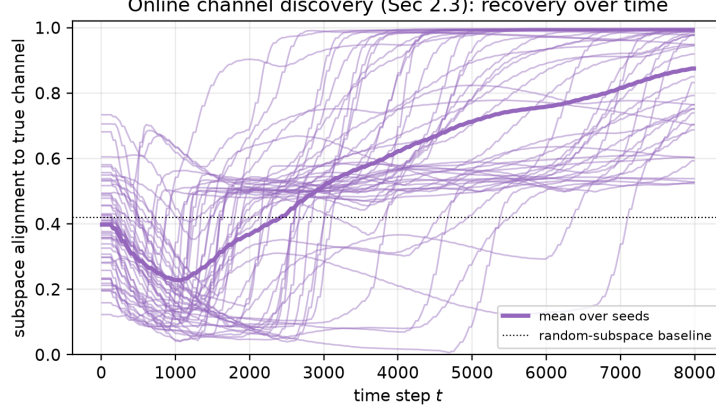


Figure 5: Online channel discovery: alignment of the learned channel  $B_t$  to the true intervention subspace over time. Thin lines are individual seeds, the bold line is their mean, and the dotted line is the random-subspace baseline.

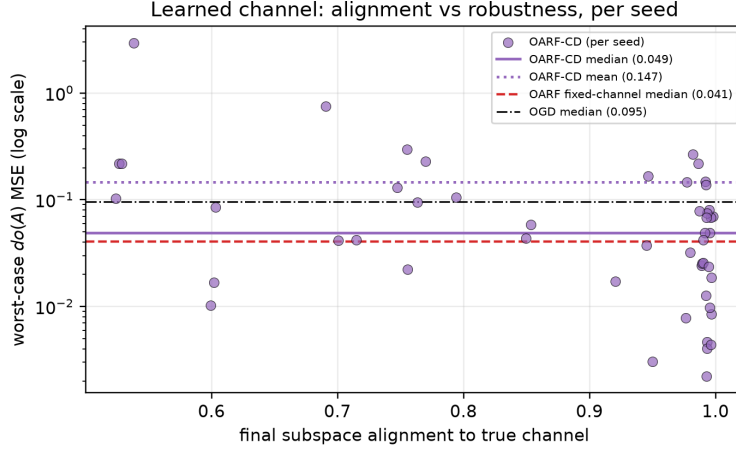


Figure 6: Per-seed alignment versus interventional robustness for the learned channel (50 seeds). Each point is one seed: the horizontal axis is the final subspace alignment of OARF-CD’s discovered channel and the vertical axis its worst-case  $\text{do}(A)$  MSE (log scale). High-alignment seeds cluster at low error near the fixed-channel OARF median (dashed), but a minority of low-alignment seeds incur an order-of-magnitude higher error. These seeds pull the mean (dotted) far above the median (solid) and make the learned channel unreliable in the worst case.

When a seed’s average post-warm-up  $\kappa$  lies in the bottom quartile (only 13 of 50 seeds), the forecaster reverts to plain OGD; otherwise it keeps the learned channel. Gating lowers the mean worst-do( $A$ ) MSE of OARF-CD from 0.147 (ungated) to 0.113, a reduction of about a quarter, and closes about two-fifths of the gap to the 0.059 attained with the true channel. The gate uses only a statistic the discovery layer already computes. Its threshold, however, is the 25th percentile of  $\kappa$  across these same 50 seeds, so in deployment it would have to be calibrated in advance, for example on simulated streams. The decision uses the average of  $\kappa$  over the second half of each stream, so the gate makes one choice per stream. It also discards a misspecified channel instead of correcting it; like Davis–Kahan, a large gap certifies a well-identified subspace but says nothing about whether that subspace is exogenous.

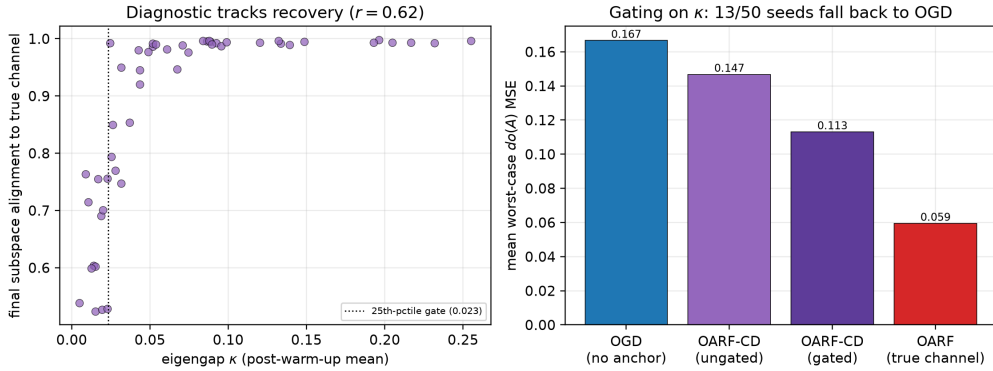


Figure 7: Eigengap diagnostic. (a) Per-seed post-warm-up eigengap  $\kappa$  vs. true subspace alignment ( $r = 0.62$ ); the dotted line is the 25th-percentile gate. (b) Mean worst-case do( $A$ ) MSE for OGD, ungated OARF-CD, OARF-CD gated on  $\kappa$ , and fixed-channel OARF with the true channel. Gating reduces the ungated learned channel’s mean worst-case error by about a quarter without using the ground truth.

*Ablations.* Table 6 and Figure 8 remove one component at a time. Most of the interventional robustness comes from the anchor channel. With the true channel the worst-case interventional error is 0.080. With a random channel it rises to 0.377, which is worse than using no anchor penalty at all (OGD, 0.226). A wrong channel therefore does active harm, so identifying the right channel is the main problem; the stress tests below measure the same effect continuously. The learned channel (0.303, alignment 0.84) is worse

than OGD on average in this experiment and, as Table 5 shows, unstable across seeds. AdaGrad matters for both accuracy and robustness: a tuned fixed step raises in-regime MSE from 0.035 to 0.095 and worst-do( $A$ ) from 0.080 to 0.145. EMA centring improves in-regime accuracy (0.035 vs. 0.043) with essentially unchanged robustness, which is consistent with separating the regime mean from the covariance the penalty uses. For the discovered dimension, worst-do( $A$ ) MSE and alignment both improve gradually as  $q$  rises from 1 to 4 ( $0.329 \rightarrow 0.261$ , alignment  $0.79 \rightarrow 0.90$ ), because a larger learned subspace still contains the true two-dimensional channel. Under-specifying the dimension ( $q = 1$ ) does more damage, because the smaller subspace cannot contain it. Results are broadly insensitive to the EMA decay  $\beta$  over  $[0.90, 0.995]$ .

Table 6: Component ablations on the synthetic benchmark (mean over 16 seeds,  $\xi = 6$  unless noted). The channel carries most of the interventional robustness, and a wrong channel is worse than none.

| Variant                                   | in-regime MSE | worst-do( $A$ ) MSE |
|-------------------------------------------|---------------|---------------------|
| OARF (full, true channel)                 | 0.035         | <b>0.080</b>        |
| – EMA centring                            | 0.043         | 0.081               |
| – AdaGrad (fixed step)                    | 0.095         | 0.145               |
| random channel                            | 0.032         | 0.377               |
| $\xi = 0$ (OGD)                           | 0.027         | 0.226               |
| learned channel (OARF-CD, alignment 0.84) | 0.033         | 0.303               |

*Stress tests.* The synthetic benchmark favours OARF by construction, because the anchor is clean and fully observed. In practice it may be neither, so we stress the fixed-channel method along two axes that break these conditions (Figure 9; the synthetic SCM is otherwise unchanged, 8 seeds). First, we add noise of growing scale to the observed context before any method sees it. Worst-do( $A$ ) MSE rises monotonically with the noise scale, and OARF crosses the no-anchor OGD baseline at roughly one standard deviation of observation noise (0.18 vs. OGD 0.19). Beyond that point a noisy anchor is no better than no anchor, and the learned channel degrades sooner. Second, we rotate the supplied channel away from the truth by an angle  $\theta$  (subspace alignment  $\cos \theta$ ). OARF beats OGD while the alignment exceeds about 0.6 (roughly  $\theta \leq 50^\circ$ ). With a more misaligned channel it is worse than

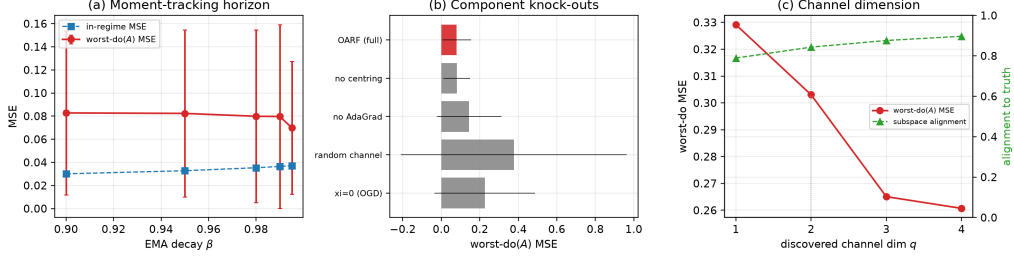


Figure 8: Ablations. (a) Sensitivity to the EMA decay  $\beta$ . (b) Component knock-outs by worst-case interventional MSE: the channel and the adaptive step matter most, and a random channel is worse than none. (c) Discovered-channel dimension: robustness and alignment both improve slightly with  $q$  (true  $q = 2$  dotted), but worst-case error stays above OGD at every  $q$ .

OGD, reaching 0.36 against OGD’s 0.19 when the channel is orthogonal; the random-channel row of Table 6 reflects the same mechanism. OARF should therefore be deployed only when the analyst has reason to believe the anchor is informative, that is, with alignment well above 0.6 and noise below order one. A weak or guessed channel can make forecasts worse than using no anchor.

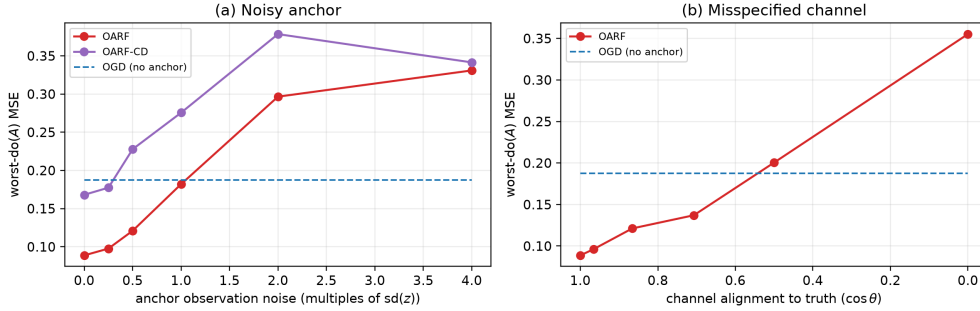


Figure 9: Stress tests of the fixed-channel method (worst-do(A) MSE; OGD baseline dashed). (a) Increasing anchor observation noise: OARF crosses the no-anchor baseline near one standard deviation of noise. (b) Rotating the channel away from the truth: anchoring helps while alignment  $\cos \theta \gtrsim 0.6$  and hurts beyond that point.

*Assumption-violation scenarios.* The stress tests vary the quality of the anchor. We also change the SCM structure so as to break the assumptions behind Proposition 1 one at a time (Table 7, median worst-do(A) MSE over

8 seeds). OARF keeps its advantage over OGD in three of these scenarios: when the intervention also shifts the parents (PARENTS), because the invariant predictor remains valid; when a second regime driver is unobserved (UNOBSERVED), because robustifying the observed part still helps; and when the observed anchor is a noisy proxy for the true latent driver (PROXY). The method therefore degrades gradually under these violations. It fails in two cases. When the regime shifts the target’s own parent-coefficient (MECHANISM), no fixed predictor is robust and both methods collapse ( $\geq 1.0$ ). When regimes shift the variance instead of the mean (VARIANCE), the mean-shift equivalence does not apply and OARF is worse than OGD. The theory covers neither case, and we regard both as limitations of the method.

Table 7: Assumption-violation scenarios (median worst-do( $A$ ) MSE over 8 seeds; lower is better). OARF uses the true observed channel. It fails under a shifting target mechanism and when regimes shift the variance instead of the mean.

| Scenario           | OARF  | OARF-CD | OGD   | assumption broken                |
|--------------------|-------|---------|-------|----------------------------------|
| clean (ideal)      | 0.021 | 0.033   | 0.051 | none (reference)                 |
| parents shifted    | 0.018 | 0.016   | 0.047 | none (still anchor-mediated)     |
| unobserved driver  | 0.024 | 0.040   | 0.048 | channel incompletely observed    |
| noisy proxy anchor | 0.044 | 0.058   | 0.065 | anchor predictive, not causal    |
| mechanism shift    | 1.299 | 1.238   | 1.020 | target mechanism non-invariant   |
| variance shift     | 0.046 | 0.040   | 0.025 | regimes shift variance, not mean |

*Comparison of anchor selectors.* Automatic anchor selection is the most fragile part of the method, so we compare the block-mean rule of OARF-CD with alternative selectors under a strict pre-deployment budget: the channel is estimated on the warm-up block only, with no look-ahead (Table 8). The alternatives are a supervised residual-context-coupling selector, stationary PCA of the context, and a random channel. Each is scored by its alignment to the truth and by the worst-do( $A$ ) MSE of the resulting fixed-channel method. The block-mean rule does about as well as the supervised selector and much better than stationary PCA. From a short warm-up, though, neither is clearly better than a random channel: their worst-do( $A$ ) MSEs are 0.101 and 0.093 against 0.115 for a random channel and 0.070 for the true one, and their alignments (0.42 and 0.37) are no higher than the random channel’s (0.43). High alignment is reached only by learning over the full

stream (OARF-CD, alignment 0.84 in Table 6), and even then the estimate is unstable. Reliable online anchor discovery from limited data thus remains an open problem. Practitioners should supply a candidate anchor and validate it with the procedure below instead of relying on automatic discovery alone.

Table 8: Anchor selectors estimated on the warm-up block only (8 seeds; alignment to the true channel, and worst-do( $A$ ) MSE of the resulting fixed-channel OARF; lower MSE / higher alignment better).

| Selector                             | subspace alignment | worst-do( $A$ ) MSE (median) |
|--------------------------------------|--------------------|------------------------------|
| true channel (reference)             | 1.00               | 0.070                        |
| block-mean (OARF-CD rule)            | 0.42               | 0.101                        |
| supervised residual-context coupling | 0.37               | 0.093                        |
| stationary PCA of context            | 0.31               | 0.236                        |
| random channel                       | 0.43               | 0.115                        |

*Diagnostics and an anchor-selection procedure.* The stress tests, scenarios and selector comparison lead to a practical procedure. Before trusting OARF, and OARF-CD in particular, a practitioner should: (1) propose candidate anchors from domain knowledge (for energy, the macro-uncertainty indices); (2) estimate the anchor-residual correlation of a plain online learner and check that it is non-trivial and stable over the warm-up; (3) run a warm-up stress test, perturbing the anchor artificially as in Figure 9, and confirm that anchoring improves validation worst-regime loss without an unacceptable in-distribution cost; (4) for OARF-CD, monitor whether the learned subspace stabilises and aligns with interpretable drivers, and fall back to a fixed anchor or to  $\xi = 0$  (plain OGD) if it does not. The checklist below summarises the resulting decision rule.

**Pre-deployment checklist for OARF.**

Use fixed-channel OARF when the anchor has a domain justification; the anchor–residual correlation is non-trivial and stable over the warm-up; a warm-up stress test shows anchoring improves validation worst-regime loss; and random-channel and noisy-anchor diagnostics do not match or beat the chosen anchor.

Do not use OARF (fall back to OGD/ACI) when the learned or chosen channel is unstable across the warm-up; validation performance is no better than plain OGD; the regime shifts appear as variance shifts or direct mechanism shifts instead of anchor-mediated mean shifts; or the anchor noise is high. Treat OARF-CD as a diagnostic only, and do not deploy its channel unless it passes these checks.

*Real data.* The energy panel (Table 9) checks whether the mechanism carries over to real series. On both targets the 10% Model Confidence Set retains 10 of the 11 methods and excludes only M-FISHER ( $p = 0.008$  on Brent). Every retained method has an MCS  $p$ -value of 1.00 except the Expected-Shortfall combiner ( $p = 0.22$  on Brent), so the differences among retained methods in Table 9 are not significant, and OARF is competitive with the strongest baselines without being better than them. The most likely reason is that real energy variance has weaker, noisier driver–residual coupling than the controlled SCM. The stress tests above predict that the anchor advantage shrinks toward zero in that setting, which is what we observe. The conformal head’s coverage is near nominal (0.90). A decision-economic volatility-timing check in the appendix finds that over this sample the risk-targeting Sharpe ratios are low and statistically indistinguishable across methods. An event study around the COVID-19 crash and the 2022 energy shock (Appendix Appendix E) gives the same picture for dated episodes. OARF holds the anchor–residual correlation below OGD’s in every window, but it has a loss advantage in only one of four event/commodity windows. The anchor mechanism is active, yet too weakly coupled to give a dependable gain.

*Significance.* The two Model Confidence Set analyses show what the evidence supports. On the synthetic benchmark, the set computed on in-distribution loss (over 16 seeds) retains the batch and conformal methods (Rolling-OLS in all 16 seeds, DRIG in 15, the conformal heads in 11) and excludes fixed-channel OARF (retained in 0 seeds) and OARF-CD (1 seed),

Table 9: Real energy log daily-variance-proxy combination, 2010–2026 evaluation window (lower is better). Ten of the eleven methods lie within the 10% Model Confidence Set (MCS  $p = 1.00$  for all but the Expected-Shortfall combiner,  $p = 0.22$  on Brent); only M-FISHER is excluded ( $p = 0.008$  on Brent). The differences among the retained methods are not statistically significant and are reported descriptively.

| Method               | Brent (DCOILBRETEU) |        |         |       | Henry Hub (DHHNGSP) |        |         |       |
|----------------------|---------------------|--------|---------|-------|---------------------|--------|---------|-------|
|                      | MSE                 | w-reg  | post-cp | cov   | MSE                 | w-reg  | post-cp | cov   |
| <b>OARF</b>          | 0.2225              | 0.2313 | 0.2384  | —     | 0.2870              | 0.3928 | 0.2670  | —     |
| <b>OARF-CD</b>       | 0.2225              | 0.2292 | 0.2433  | —     | 0.2857              | 0.3850 | 0.2627  | —     |
| OGD                  | 0.2243              | 0.2334 | 0.2412  | —     | 0.2860              | 0.3888 | 0.2655  | —     |
| Rolling-OLS          | 0.2203              | 0.2230 | 0.2358  | —     | 0.2882              | 0.3916 | 0.2623  | —     |
| ACI                  | 0.2180              | 0.2221 | 0.2399  | 0.900 | 0.2808              | 0.3833 | 0.2426  | 0.904 |
| M-FISHER (2025)      | 0.2680              | 0.2691 | 0.3307  | —     | 0.3532              | 0.4428 | 0.3442  | —     |
| DR-Combo (Wang 2026) | 0.2304              | 0.2325 | 0.2507  | —     | 0.2891              | 0.3958 | 0.2432  | —     |
| FC-DRO (Liu 2026)    | 0.2318              | 0.2346 | 0.2571  | —     | 0.2872              | 0.3916 | 0.2436  | —     |
| FC-DRO-ES (Liu 2026) | 0.2387              | 0.2440 | 0.2713  | —     | 0.3056              | 0.4086 | 0.2689  | —     |
| W-DRO-OL (Chen 2026) | 0.2243              | 0.2334 | 0.2406  | —     | 0.2859              | 0.3890 | 0.2633  | —     |
| AnchorReg            | 0.2183              | 0.2195 | 0.2345  | —     | 0.2907              | 0.3942 | 0.2716  | —     |

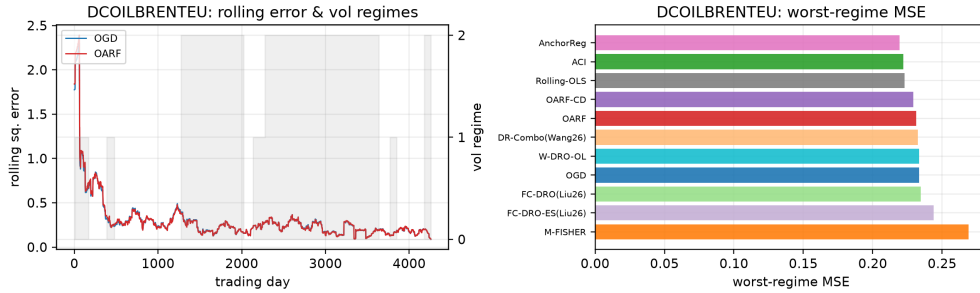


Figure 10: Brent: rolling squared error with the volatility-regime overlay (left) and worst-regime MSE by method (right).

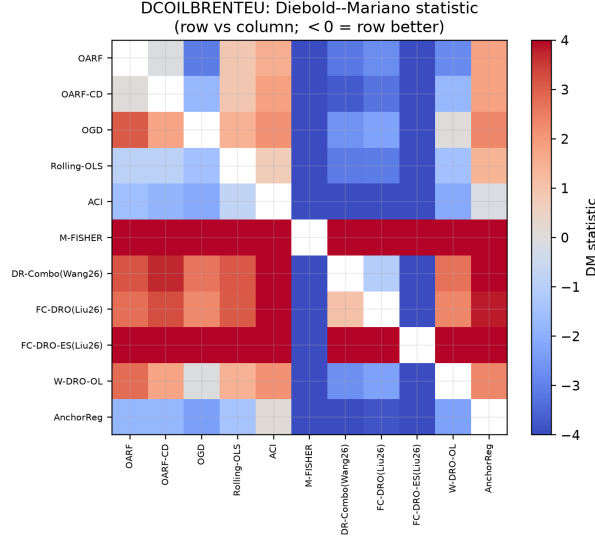


Figure 11: Brent: pairwise Diebold–Mariano statistics (negative = row more accurate than column). Most pairs are small in magnitude, consistent with the Model Confidence Set retaining ten of the eleven methods.

because OARF gives up some in-distribution accuracy in exchange for interventional robustness, which in-distribution loss does not reward; the relevant evaluation for OARF is the held-out intervention grid. On the real targets the MCS retains 10 of 11 methods on each commodity (only the weakest is excluded), so no method significantly out-predicts the others on average, and the real-data differences are descriptive, as noted above. The Diebold–Mariano statistics agree: they are pronounced on the synthetic benchmark (Figure 12) and mostly small on the real targets (Figure 11).

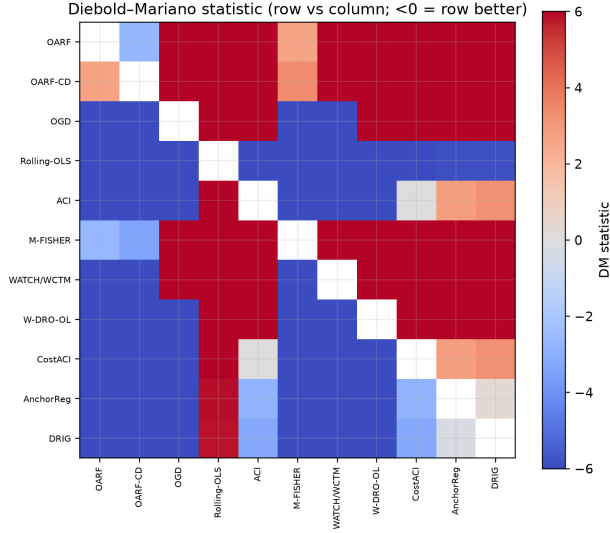


Figure 12: Pairwise Diebold–Mariano statistics on the synthetic benchmark; a negative entry means the row method is significantly more accurate than the column method.

## 6. Discussion and conclusion

When regime shifts pass through an observable channel with anchor-mediated confounding, a forecaster can be made robust to them in advance and online. The gain is a large reduction in worst-case interventional error, paid for by an increase in in-distribution error that is real but can be tuned. When that structure is weak, as on the real energy series, the method is competitive with the strongest baselines but not significantly better. Geometrically, distributionally robust optimisation spreads its protection isotropically, while anchoring concentrates it on the directions in which the regime moves. The concentration helps only to the extent that those directions are correctly identified.

*Scope.* The mechanism that helps under a good anchor also hurts under a bad one. OARF is likely to help when the anchor is close to exogenous, regimes are mainly mean shifts in the anchor, the anchor is observed with at most moderate noise, and its alignment with the true channel is above about 0.6. It is likely to do worse than plain OGD when the anchor is misspecified or weakly aligned, when shifts act on the target mechanism or on the variance of the anchor, when the learned channel is unstable, or when the stream is

too short or regimes change faster than the EMA window can track. The checklist in Section 5 turns these conditions into a pre-deployment test, and for the learned channel the eigengap  $\kappa$  of Eq. (6) gives an online check; in our simulations reverting to OGD on the streams it flags cut the learned channel’s mean worst-case error by about a quarter, with a threshold set on the same simulations.

*Conclusion.* Online anchor regression with a fixed channel gives proactive robustness to regime shifts, backed by a fixed-anchor interventional-regret guarantee. This holds only when the anchor behaves like an exogenous variable, is informative and is observed with moderate noise, and only for mean-shift regimes. Outside those conditions the method brings no benefit and can do harm. On sixteen years of daily energy variance-proxy forecasting, ten of the eleven methods lie in the Model Confidence Set, which is what our stress tests predict for a weak, noisy channel.

Online discovery of the anchor channel remains open. The block-mean rule is reasonable but unstable, and on short data no selector does clearly better than a random channel. We made progress on a narrower question that matters more for deployment: knowing when not to trust the discovered channel. The eigengap diagnostic of Eq. (6) is a partial answer. It flags unreliable recovery without certifying that a well-identified subspace is exogenous. Section 5 reports the scenarios it cannot help with (target-mechanism and variance shifts), where no channel, learned or supplied, works. In practice, OARF should be used with diagnostics: propose a candidate anchor, validate it on a warm-up stress test, and fall back to plain online learning when the anchor, or the stability diagnostic of the discovered channel, shows that it is weak or unstable.

The contributions are (i) a streaming form of anchor regression whose updates use only past data with a fixed-anchor interventional-regret guarantee; (ii) an evaluation protocol for online forecasting based on interventional regret; (iii) a diagnostic that needs no ground truth and lets a deployed system detect an unreliable discovered channel and revert to plain online learning, which limits the worst-case cost of not knowing the channel; and (iv) an empirical study of when anchor-based proactive robustness helps or fails, with a pre-deployment checklist for deciding whether it is appropriate for a given problem. Future work includes tightening the link between the eigengap and a formal guarantee of robustness to misspecification, sharper online channel discovery, nonlinear anchors and predictors, richer interventions on

variances and mechanisms, commodity-specific implied-volatility data, and event-triggered monitoring in deployment.

*Open theoretical question.* The EMA-moment rate of  $T^{2/3}$  in the fixed-channel theorem rests, through Lemma 3, on stationarity within regimes (Assumption 4). The main open problems are to tighten this rate toward  $\sqrt{T}$  without a changepoint trigger and to extend a regret guarantee to the learned or misspecified channel. The eigengap diagnostic of Eq. (6) is a step in this direction, since Davis–Kahan already supplies the estimation-error half of such an argument. Connecting it to a regret bound, beyond the empirical validation reported here, remains open.

*Reproducibility.* All code and data-build scripts that regenerate every number, table and figure in this paper will be released in a public repository upon acceptance. For double-anonymized review the repository link is withheld from this anonymized manuscript; it is given to the editor on the title page.

## Appendix A. Proof of Theorem 1

Throughout,  $\mathcal{R}_t(w) := \sup_{\|v\| \leq \rho(\xi)} \mathbb{E}_{\text{do}(a:=\nu)}[(y_t - w^\top \phi(x_t))^2]$  is the per-step worst-case risk in (7), and  $\hat{g}_t$  is the (inexact) step direction of Algorithm 1.

**Lemma 1** (Convex surrogate and gradient decomposition).  *$\mathcal{R}_t$  is convex, and under Assumption 2 it equals, within the active regime, the anchor-penalised risk  $\mathcal{R}_t(w) = \mathbb{E}_t[(y - w^\top \phi)^2] + \xi P_t(w)$  of (1). Moreover  $\hat{g}_t = \nabla \mathcal{R}_t(w_t) + \zeta_t$  with  $\zeta_t = M_t + b_t$ , where  $M_t := -r_t \phi(x_t) - \mathbb{E}_t[-r\phi]$  is a martingale difference ( $\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = 0$ ) and  $b_t$  is the deterministic ( $\mathcal{F}_{t-1}$ -measurable) error of the EMA-estimated penalty gradient relative to  $\nabla(\xi P_t)$ .*

*Proof.* For fixed  $\nu$ ,  $\mathbb{E}_{\text{do}(a:=\nu)}[(y - w^\top \phi)^2]$  is a convex quadratic in  $w$ ; a point-wise supremum of convex functions is convex. The identity is Proposition 1 within the stationary active regime. The prediction term  $-r_t \phi(x_t)$  is an unbiased one-sample estimate of  $\frac{1}{2} \nabla \mathbb{E}_t[(y - w^\top \phi)^2] = -\mathbb{E}_t[r\phi]$ , with centred part  $M_t$ ; the robustness term substitutes the  $\mathcal{F}_{t-1}$ -measurable EMA moments for the population moments in  $\nabla(\xi P_t)$ , and  $b_t$  is that substitution error.  $\square$

**Lemma 2** (Inexact dynamic regret). *Run projected OGD with the path-length-adaptive Ader wrapper (Zhang et al., 2018) on the convex losses  $\mathcal{R}_t$ , fed  $\hat{g}_t$ . Then for any  $u_{1:T} \subset \mathcal{W}$  of path length  $P_T$ ,  $\sum_t \mathcal{R}_t(w_t) - \sum_t \mathcal{R}_t(u_t) \leq$*

$O(\sqrt{T(1+P_T)}) + \sum_t \langle \zeta_t, w_t - u_t \rangle$ , with the first term attained without prior knowledge of  $P_T$ .

*Proof.* With exact gradients, Ader attains minimax dynamic regret  $O(\sqrt{T(1+P_T)})$  for convex  $G$ -Lipschitz losses over a radius- $D$  set, adaptively in  $P_T$  (Zhang et al., 2018), subsuming Zinkevich (2003); Hazan (2016). Convexity gives  $\mathcal{R}_t(w_t) - \mathcal{R}_t(u_t) \leq \langle \nabla \mathcal{R}_t(w_t), w_t - u_t \rangle = \langle \hat{g}_t, w_t - u_t \rangle - \langle \zeta_t, w_t - u_t \rangle$ ; summing and applying the exact-gradient bound to the  $\hat{g}_t$  terms yields the inexactness term.  $\square$

**Lemma 3** (Gradient-error control). *Under Assumptions 1–4, against the deterministic minimising comparator: (i)  $\mathbb{E}[\sum_t \langle M_t, w_t - u_t \rangle] = 0$ ; and (ii)  $\mathbb{E}\|b_t\| \leq c_1\sqrt{1-\beta}$  at steps at least  $1/(1-\beta)$  into a regime, with  $\|b_t\| \leq c_2$  within  $1/(1-\beta)$  steps of each of the  $K$  changes, so  $\sum_t \mathbb{E}\|b_t\| \leq c_1T\sqrt{1-\beta} + c_2K/(1-\beta)$ ; with oracle moments  $b_t \equiv 0$ .*

*Proof.* (i)  $w_t$  is  $\mathcal{F}_{t-1}$ -measurable and, the schedule being oblivious (Assumption 3),  $u_t$  is deterministic, so  $\mathbb{E}[\langle M_t, w_t - u_t \rangle \mid \mathcal{F}_{t-1}] = 0$ ; the tower rule concludes. (ii) An EMA of decay  $\beta$  of a stationary sequence has variance  $\Theta(1-\beta)$  about its mean and geometrically-decaying within-regime bias; by Assumption 1 the penalty gradient is Lipschitz in these moments (the  $\epsilon$ -ridge bounds the inverse), inheriting the  $\Theta(\sqrt{1-\beta})$  steady-state scale. After a change the EMA mixes the previous regime for  $O(1/(1-\beta))$  steps with bounded  $\|b_t\| \leq c_2$ . Summing gives the bound; oracle moments make the substitution error vanish.  $\square$

*Proof of Theorem 1.* Combining the lemmas and taking expectations, for the deterministic minimiser  $u_{1:T} \in \mathcal{W}$  ( $P_T \leq 2DK$ ),  $\mathbb{E}[\sum_t \mathcal{R}_t(w_t)] - \sum_t \mathcal{R}_t(u_t) \leq O(\sqrt{T(1+K)}) + 0 + D\sum_t \mathbb{E}\|b_t\|$ . With oracle moments the last term vanishes, giving (i); with EMA moments it is  $\leq D(c_1T\sqrt{1-\beta} + c_2K/(1-\beta))$ , which at  $1-\beta_T = T^{-2/3}$  is  $O((1+K)T^{2/3})$  and dominates  $\sqrt{T(1+K)}$ , giving (ii). Finally, by Assumption 2 the realised regime is an admissible intervention, so  $\mathbb{E}_t[\ell_t(w_t) \mid \mathcal{F}_{t-1}] \leq \mathcal{R}_t(w_t)$  and hence  $\mathbb{E}[\sum_t \ell_t(w_t)] \leq \mathbb{E}[\sum_t \mathcal{R}_t(w_t)]$ ; as  $\min_{u \in \mathcal{W}} \sum_t \mathcal{R}_t(u)$  is the comparator in (7), the displays bound  $\mathbb{E}[\text{IntReg}_T(\rho(\xi))]$  by the stated rates.  $\square$

## Appendix B. Decision-economic evaluation (secondary)

As a secondary check we convert each method’s volatility forecast into a volatility-timing portfolio (Fleming et al., 2001). Exposure is scaled inversely

to the forecast variance,  $w_t = \text{clip}(\sigma^{*2}/\hat{\sigma}_t^2, 0, 3)$ , and the strategy earns  $w_t r_{t+1}$  net of a one-basis-point turnover cost. We report the annualised Sharpe ratio, maximum drawdown and turnover. Over 2010–2026 the Sharpe ratios of these commodity risk-targeting strategies are low and statistically indistinguishable across methods (Figure B.13). The evaluation is reported for completeness only; it carries no evidential weight, and no claim in the main text rests on it.

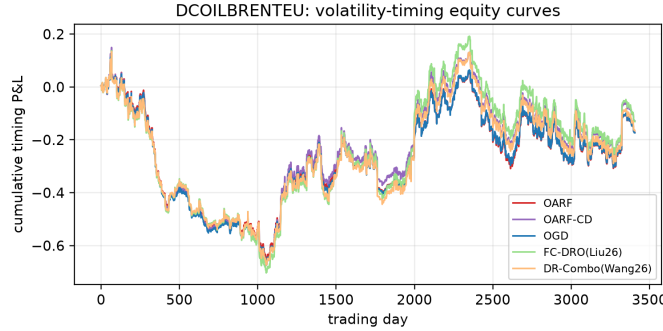


Figure B.13: Brent: cumulative volatility-timing profit and loss. Differences across methods are economically small and not statistically significant over this sample.

## Appendix C. Baseline implementations and notes

Table C.10 lists, for each method, the loss optimised, the update rule and the output, whether the method runs online, rolling or in batch, and whether it uses the anchor. All methods run under the same protocol with the hyper-parameters of Table 4. Per step, the linear methods cost  $O(d)$  and OARF adds  $O(pq + q^3)$  for the anchor block. Rolling-OLS and the batch methods cost  $O(d^3)$  at each refit, which takes place every 5–10 steps.

Some implementation details follow. The warm-up block is used only to initialise the past-only standardiser and to select hyper-parameters once; it is never used for separate batch pre-training. Missing FRED observations and exchange holidays are handled by reindexing to business days with a forward-fill capped at five days, and longer gaps are left missing and skipped. After the warm-up the GARCH(1,1) expert keeps fixed RiskMetrics-style coefficients and is not re-estimated online, so it is a fixed-parameter recursion. The channel dimension is larger on the real data ( $q = 3$ , against  $q = 2$  in the synthetic experiments) because three macro-uncertainty drivers (VIX,

GPR, EPU) are chosen by hand as the anchor, while the synthetic SCM has exactly two anchor directions by construction. We present OARF for linear predictors. It extends to nonlinear forecasters by applying the same residual–anchor penalty to a feature map or to the top layer of a network; we leave this extension to future work.

Table C.10: Baseline implementations. “Out” = P(oint)/I(nterval); “Mode” = O(nline)/R(olling)/B(atch); “Anc” = uses the anchor.

| Method       | Loss optimised             | Update rule           | Out | Mode | Anc   |
|--------------|----------------------------|-----------------------|-----|------|-------|
| OARF/OARF-CD | anchor-penalised sq. err.  | AdaGrad on (4)        | P   | O    | yes   |
| OGD          | squared error              | AdaGrad ( $\xi = 0$ ) | P   | O    | no    |
| Rolling-OLS  | windowed ridge             | closed-form refit     | P   | R    | no    |
| ACI/CostACI  | pinball (3 quantiles)      | subgrad. + ACI level  | P+I | O    | no    |
| M-FISHER     | sq. err. + martingale      | OGD + triggered NGD   | P   | O    | no    |
| WATCH/WCTM   | sq. err. + WCTM            | OGD + width adapt     | P+I | O    | no    |
| DR-Combo     | Wasserstein-DR combo       | EG mirror descent     | P   | O    | no    |
| FC-DRO(-ES)  | variance-/ES-scaled        | exponential weights   | P   | O    | no    |
| W-DRO-OL     | Wasserstein-DR sq. err.    | primal–dual OGD       | P   | O    | no    |
| AnchorReg    | anchor-penalised (batch)   | windowed closed-form  | P   | R    | yes   |
| DRIG         | invariant-gradient (batch) | windowed two-env. fit | P   | R    | (env) |

## Appendix D. Ader-wrapped versus AdaGrad OARF

The interventional-regret theorem (Theorem 1) is stated for the path-length-adaptive Ader step wrapper (Zhang et al., 2018), whereas the deployed model takes a single AdaGrad step. To check whether this simplification changes the empirical behaviour, we implemented the Ader variant and compared it with the deployed AdaGrad model on the controlled anchor-SCM (true fixed channel,  $\xi = 6, 12$  seeds). The Ader variant is a pool of five gradient-descent experts that follow the same anchor-robust surrogate gradient with a geometric ladder of step sizes. Each expert is projected onto the comparator ball, and a Hedge meta-learner combines them. Table D.11 reports the results. Both step rules cut the worst-case  $\text{do}(A)$  error well below the no-anchor OGD reference, to about a third (AdaGrad) and a half (Ader) of it, and their per-seed worst-case errors are strongly rank-correlated (Spearman  $\rho = 0.80$ ). In this experiment AdaGrad is slightly more accurate and has lower variance. It also has fewer parameters, and for these reasons it

is the rule we deploy; the Ader wrapper is the object the theorem analyses. Substituting one for the other leaves the qualitative ranking unchanged.

Table D.11: Ader-wrapped versus AdaGrad OARF on the synthetic anchor-SCM (12 seeds, true fixed channel,  $\xi = 6$ ; median with  $[q_{25}, q_{75}]$ ). Both reach similar worst-case interventional error, far below the OGD reference, and their per-seed worst-case errors are rank-correlated at Spearman  $\rho = 0.80$ .

| Step rule                    | in-regime MSE | worst-do( $A$ ) MSE  |
|------------------------------|---------------|----------------------|
| AdaGrad (deployed)           | 0.028         | 0.052 [0.030, 0.114] |
| Ader (analysed)              | 0.030         | 0.073 [0.026, 0.142] |
| OGD ( $\xi = 0$ , reference) | 0.019         | 0.143 [0.065, 0.353] |

## Appendix E. Event study: COVID-19 and the 2022 energy shock

The regime-tercile analysis in the main text is aggregate, so here we examine two dated episodes. We run fixed-anchor OARF ( $\xi = 1.5$ ) and the no-anchor OGD reference on the real combination stream. Within each event window we plot the cumulative squared-loss gap (OARF minus OGD, so a negative value means OARF is ahead) and, for both methods, the smoothed anchor-residual correlation  $|\text{corr}(a_t, r_t)|$  (Figure E.14). In every window OARF keeps the anchor-residual correlation visibly below that of OGD, so on real data the update does what it is designed to do. This does not produce a consistent loss advantage. OARF is ahead during the COVID-19 window on Brent (cumulative squared-loss gap  $-0.27$ ) and behind in the other three window/commodity combinations (gaps of  $+0.30$  to  $+1.58$ ). The pattern matches the Model-Confidence-Set result: on this public daily data the anchor mechanism is active, but it is too weakly coupled to the target to give a dependable forecasting edge. The event study illustrates this around recognisable episodes; it does not demonstrate that OARF is superior.

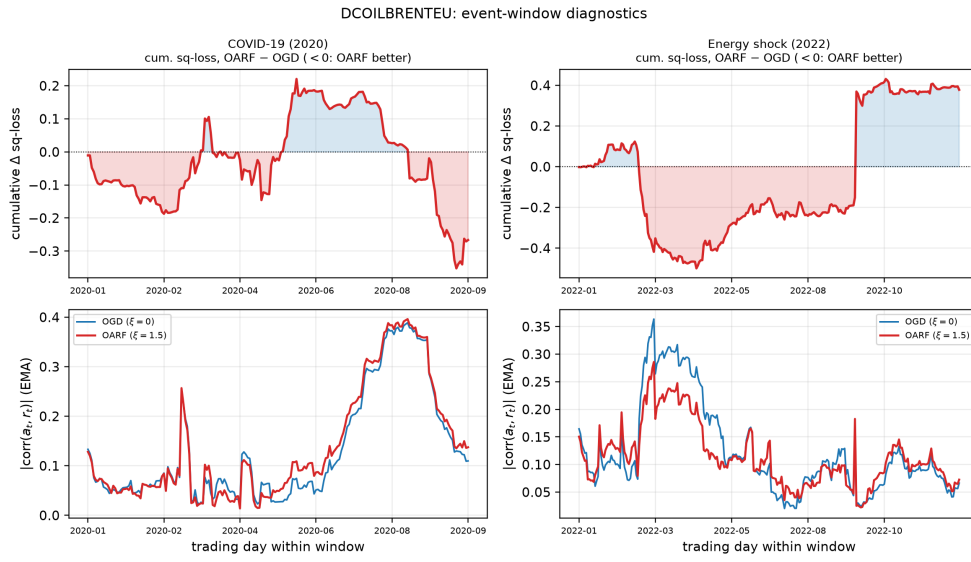


Figure E.14: Brent event windows. Top: cumulative squared-loss gap, OARF minus OGD (shaded blue where OARF leads, orange where it trails). Bottom: the smoothed anchor-residual correlation, which OARF holds below OGD in both windows. OARF leads during the 2020 COVID-19 crash but trails during the 2022 energy shock, consistent with the statistically insignificant aggregate result.

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## **CRedit authorship contribution statement**

The corresponding author is responsible for conceptualization, methodology, software, formal analysis, investigation, data curation, writing (original draft, review and editing), and visualization.

## **Data availability**

Public data sources and the code that regenerates every number, table and figure in this paper (see Reproducibility, above) will be made available in a public repository upon acceptance.

## **Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**

The author used Claude (Anthropic) between June and September 2026 to help write the analysis code and to edit the text of the manuscript. The author reviewed and revised all code and text and takes full responsibility for the content of this article.

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